



End Term (Even) Semester Regular/Back Examination May 2026

Roll no. **2294 038**

Name of the Program and semester: B.Tech 8th sem (CSE)
Name of the Course : Wind and solar energy systems
Course Code: TOE805

Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1.

(2X10=20 Marks)

a. (i) Compare and contrast conventional and non-conventional energy resources. Discuss why the global energy scenario is rapidly shifting toward the latter. (5 Marks) CO1

(ii) Explain the concept of Sustainable Development in the context of energy production. How does the widespread adoption of renewable energy mitigate the specific environmental impacts caused by fossil fuels? (5 Marks) CO1

b. Explain solar radiation and its components. Describe the instruments used for measuring solar radiation such as pyranometer and pyrheliometer with neat diagrams. CO2

c. Explain the current-voltage (I-V) characteristics of a solar cell. Discuss the significance of short circuit current, open circuit voltage, and fill factor. CO3

Q2.

(2X10=20 Marks)

a. Explain how wind characteristics, turbine design, and site selection together influence the efficiency and performance of a wind energy system. CO4

b. Explain the concept of power extraction from wind. Derive the Betz limit and discuss why it sets the maximum efficiency of a wind turbine. CO3

c. Discuss the regional variations in energy resources and consumption. Explain how geography and development affect energy distribution. CO1

Q3.

(2X10=20 Marks)

a. Discuss various applications of solar thermal energy such as water heating, drying, cooking, and power generation. Explain any one in detail. CO2

b. Explain methods of wind data collection and analysis. Discuss wind rose diagram, Weibull distribution, and their significance. CO4

c. Discuss the economic feasibility of wind energy projects. Explain factors such as initial cost, maintenance, payback period, and government incentives. CO5

Q4.

(2X10=20 Marks)

a. Explain the working principle of a Wind Energy Conversion System (WECS) with a neat block diagram. Describe the energy conversion process from wind to electrical energy. CO5



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b. Explain the key factors in designing a PV system such as load estimation, tilt angle, orientation, shading, and battery sizing. CO3

c. Describe different types of concentrating collectors. Explain their working principle and compare them with flat plate collectors. CO2

Q5. (2X10=20 Marks)

a. Explain the aerodynamic principles involved in wind turbines. Discuss lift and drag forces and their role in energy conversion. CO4

b. Analyze the environmental impacts of wind energy such as noise, visual impact, and effects on wildlife. Suggest mitigation measures. CO5

c. A solar energy system is to be installed at a given location. Explain how solar geometry (declination, hour angle, altitude, azimuth) affects the design and efficiency of the system. CO2